Supplementary material

Hospitalizations for ambulatory care sensitive conditions in Ecuador, 2000–2024. Martínez-Abarca AL, Henríquez-Trujillo MC, Bonilla E, Henríquez-Trujillo AR. Data and code: https://doi.org/10.5281/zenodo.23366479.

Tables S4–S9 are provided as spreadsheet files: S4 joinpoint regression of all series; S5 segmented regression by ACSC group; S6 sensitivity analyses; S7 annual provincial age-standardized rates with 95% confidence intervals for 2000–2024; S8 annual indicators of geographic variation; S9 hospital bed supply by province and its association with hospitalizations, 2004–2024.

Table S1. ICD-10 codes of the 20 groups of ambulatory care sensitive conditions (ACSC), number of hospitalizations and age-standardized rates by period, Ecuador, 2000–2024.

Group	Condition	ICD-10 codes	Category	n 2000–2024	% of ACSC	ASR 2000–2007	ASR 2008–2014	ASR 2015–2019	ASR 2020–2024
1	Anemia	D50	Other acute	19,951	0.5	2.4	5.0	8.2	6.7
2	Angina pectoris	I20	Chronic	39,306	1.0	10.6	12.1	10.7	13.7
3	Asthma	J45-J46	Chronic	70,906	1.9	22.9	20.2	14.4	11.4
4	Nutritional deficiencies	E40-E46, E50-E64	Other acute	27,745	0.7	9.1	7.0	5.0	3.6
5	Diabetes mellitus	E10-E14	Chronic	357,721	9.4	100.8	138.4	108.5	83.4
6	Female pelvic inflammatory disease	N70-N73, N75-N76	Other acute	73,792	1.9	21.2	23.0	21.2	14.6
7	Cerebrovascular diseases	I63-I67, I69, G45-G46	Chronic	174,416	4.6	55.8	59.4	51.4	45.6
8	Lower respiratory tract diseases	J20-J21, J40-J44, J47	Chronic	214,413	5.7	49.6	70.0	64.7	40.2
9	Immunization-preventable diseases	A17.0, A19, A33-A37, A95, B05-B06, B16, B26, G00.0	Vaccine-preventable	11,135	0.3	3.6	2.8	2.8	1.7
10	Conditions related to prenatal and childbirth care	A50, O23, P35.0	Infectious	253,630	6.7	42.3	69.0	83.7	67.7
11	Epilepsy	G40-G41	Other acute	80,692	2.1	12.2	19.3	26.9	25.6
12	Infectious gastroenteritis and complications	A00-A09, E86	Infectious	990,268	26.1	304.3	290.3	204.1	128.8
13	Hypertension	I10-I11	Chronic	178,021	4.7	71.7	73.3	41.8	29.4
14	Ear, nose and throat infections	H66, J00-J03, J06, J31	Infectious	103,681	2.7	22.0	29.4	28.2	19.9
15	Skin and subcutaneous tissue infections	A46, L01-L04, L08	Infectious	230,020	6.1	45.8	62.6	71.9	60.9
16	Kidney and urinary tract infections	N10-N12, N30, N34, N39.0	Infectious	453,185	11.9	94.0	142.4	145.3	107.5
17	Heart failure	I50, J81	Chronic	120,795	3.2	45.3	41.5	32.1	28.5
18	Bacterial pneumonia	J13-J14, J15.3-J15.4, J15.8-J15.9, J18.1	Vaccine-preventable	125,140	3.3	15.1	23.8	53.8	40.2
19	Gastrointestinal ulcer	K25-K28, K92.0-K92.2	Other acute	161,981	4.3	46.2	50.6	46.7	47.4
20	Other avoidable conditions	A15-A16, A17.1-A17.9, A18, A51-A53, B50-B54, B77, I00-I02	Vaccine-preventable	105,762	2.8	64.6	20.2	13.9	12.0

Main diagnosis only. Three-character items include all subcategories; four-character items do not include records coded only with the three-character category. ASR: age-standardized rate per 100,000, projected 2022 population of Ecuador as standard.

Table S2. Age-standardized ACSC hospitalization rates by province of residence and period, Ecuador, 2000–2024.

		Age-standardized rate per 100,000						
Province	**Region**	**2000–2007**	**2008–2014**	**2015–2019**	**2020–2024**	**2000–2024**	**APC 2011–2019, % (95%CI)**	**LISA**
Morona Santiago	Amazon	1,910.0	2,457.5	2,508.0	1,787.6	2,162.4	−1.4 (−3.9 to 1.3)	HH
Pastaza	Amazon	2,613.5	2,237.8	1,790.0	1,184.2	1,920.6	−1.5 (−3.5 to 0.5)	HH
Napo	Amazon	2,032.1	2,045.4	1,418.0	1,039.6	1,626.4	−4.8 (−7.9 to −1.5)	NS
Zamora Chinchipe	Amazon	1,332.1	1,694.5	1,757.3	1,549.2	1,571.8	2.1 (0.7 to 3.5)	NS
Cañar	Highlands	1,265.4	1,560.7	1,521.1	1,186.7	1,376.9	0.0 (−1.9 to 1.9)	NS
Galápagos	Galápagos	954.8	1,384.0	1,611.9	928.7	1,226.5	−0.1 (−3.6 to 3.6)	–
El Oro	Coast	1,237.9	1,382.6	1,344.6	851.1	1,193.7	−1.7 (−4.7 to 1.4)	NS
Azuay	Highlands	1,189.8	1,268.8	1,106.1	882.4	1,121.4	−2.2 (−3.4 to −0.9)	NS
Tungurahua	Highlands	980.3	1,337.5	1,202.8	951.5	1,112.6	−2.6 (−4.8 to −0.3)	LH
Los Ríos	Coast	1,314.5	1,276.3	1,134.8	717.9	1,108.2	−2.9 (−4.8 to −1.0)	NS
Santo Domingo de los Tsáchilas	Coast	1,026.1	1,132.2	1,087.4	972.9	1,052.6	0.2 (−2.5 to 3.1)	NS
Chimborazo	Highlands	948.1	1,176.5	1,184.5	870.2	1,038.8	−0.5 (−1.1 to 0.2)	NS
Esmeraldas	Coast	1,267.6	1,082.2	948.2	790.4	1,023.3	0.4 (−3.1 to 4.1)	NS
Manabí	Coast	1,001.8	1,274.7	950.9	795.3	1,011.4	−6.0 (−8.3 to −3.7)	NS
Loja	Highlands	887.5	1,217.4	1,064.9	853.4	1,008.0	−2.7 (−4.9 to −0.4)	NS
Sucumbíos	Amazon	890.3	1,066.8	1,247.1	756.5	987.5	0.5 (−4.0 to 5.2)	NS
Cotopaxi	Highlands	989.3	1,184.8	1,107.3	646.0	980.8	−2.0 (−3.6 to −0.3)	NS
Santa Elena	Coast	774.6	1,070.4	1,217.9	752.0	948.5	−0.9 (−4.0 to 2.3)	NS
Carchi	Highlands	934.1	972.4	961.9	876.3	941.1	0.0 (−1.9 to 2.0)	NS
Imbabura	Highlands	877.6	1,101.1	867.9	882.0	935.9	−3.0 (−5.0 to −0.9)	NS
Pichincha	Highlands	1,002.5	1,003.2	964.7	809.7	927.1	0.1 (−0.8 to 1.0)	NS
Guayas	Coast	1,001.1	1,067.0	893.7	649.4	906.9	−3.0 (−4.7 to −1.4)	NS
Bolívar	Highlands	727.8	974.1	951.7	733.5	839.4	−1.4 (−2.3 to −0.5)	NS
Orellana	Amazon	726.3	1,018.3	837.0	625.7	800.3	−3.6 (−6.1 to −1.1)	NS
Geographic variation								
Extremal quotient		3.60	2.53	3.00	2.86	2.70		
Coefficient of variation, %		38.3	30.0	30.9	29.8	29.4		
Systematic component of variation		16.6	10.8	13.4	12.8	11.1		
Moran's I, contiguity (p)		0.08 (0.103)	0.26 (0.012)	0.24 (0.013)	0.30 (0.006)	0.24 (0.017)		
Moran's I, 4 nearest (p)		0.02 (0.253)	0.13 (0.072)	0.10 (0.103)	0.08 (0.139)	0.09 (0.114)		

Rates per 100,000 person-years. APC: annual percentage change 2011–2019 from a log-linear model. LISA: local Moran's I cluster for 2000–2024 at $p < 0.05$, queen contiguity (HH high-high; LH low-high; NS not significant; – no neighbours). Moran's I with 999 permutations; Galápagos excluded from contiguity statistics.

Table S3. Five most frequent ACSC diagnoses by age group and sex, Ecuador, 2000–2024.

Rank	ICD-10	Diagnosis (INEC label)	n	% of ACSC in stratum	Crude rate per 100,000
0-14 years, men					
1	A09	Diarrea y gastroenteritis de presunto origen infeccioso	233,232	38.7	389.8
2	J20	Bronquitis aguda	33,397	5.5	55.8

Rank	ICD-10	Diagnosis (INEC label)	n	% of ACSC in stratum	Crude rate per 100,000
3	J15.9	Neumonía bacteriana, no especificada	30,167	5.0	50.4
4	L02	Absceso cutáneo, furúnculo y ántrax	24,224	4.0	40.5
5	E86	Depleción del volumen	23,989	4.0	40.1
0-14 years, women					
1	A09	Diarrea y gastroenteritis de presunto origen infeccioso	193,703	34.6	337.4
2	N39.0	Infección de vías urinarias, sitio no especificado	60,633	10.8	105.6
3	J20	Bronquitis aguda	26,438	4.7	46.0
4	J15.9	Neumonía bacteriana, no especificada	26,154	4.7	45.6
5	E86	Depleción del volumen	21,466	3.8	37.4
15-64 years, men					
1	A09	Diarrea y gastroenteritis de presunto origen infeccioso	76,408	12.4	63.3
2	E11	Diabetes mellitus no insulinodependiente	46,446	7.5	38.5
3	K92.2	Hemorragia gastrointestinal, no especificada	36,426	5.9	30.2
4	E14	Diabetes mellitus, no especificada	32,485	5.3	26.9
5	I10	Hipertensión esencial (primaria)	30,596	5.0	25.4
15-64 years, women					
1	O23	Infección de las vías genitourinarias en el embarazo	245,249	21.9	200.3
2	N39.0	Infección de vías urinarias, sitio no especificado	126,655	11.3	103.4
3	A09	Diarrea y gastroenteritis de presunto origen infeccioso	100,706	9.0	82.2
4	E11	Diabetes mellitus no insulinodependiente	54,385	4.9	44.4
5	N12	Nefritis tubulointersticial, no especificada como aguda o crónica	54,134	4.8	44.2
65+ years, men					
1	I50	Insuficiencia cardíaca	39,422	9.6	353.5
2	J44	Otras enfermedades pulmonares obstructivas crónicas	33,438	8.2	299.9
3	I10	Hipertensión esencial (primaria)	31,590	7.7	283.3
4	K92.2	Hemorragia gastrointestinal, no especificada	30,652	7.5	274.9
5	E11	Diabetes mellitus no insulinodependiente	30,387	7.4	272.5
65+ years, women					
1	I10	Hipertensión esencial (primaria)	46,803	9.7	346.8
2	E11	Diabetes mellitus no insulinodependiente	46,282	9.6	343.0
3	N39.0	Infección de vías urinarias, sitio no especificado	42,057	8.7	311.7
4	I50	Insuficiencia cardíaca	39,770	8.2	294.7
5	A09	Diarrea y gastroenteritis de presunto origen infeccioso	34,254	7.1	253.8

Three-character ICD-10 categories, except where the ACSC list includes only one subcategory (J15.9, N39.0, K92.2). Diagnosis labels as published by INEC. Crude rates per 100,000 person-years.

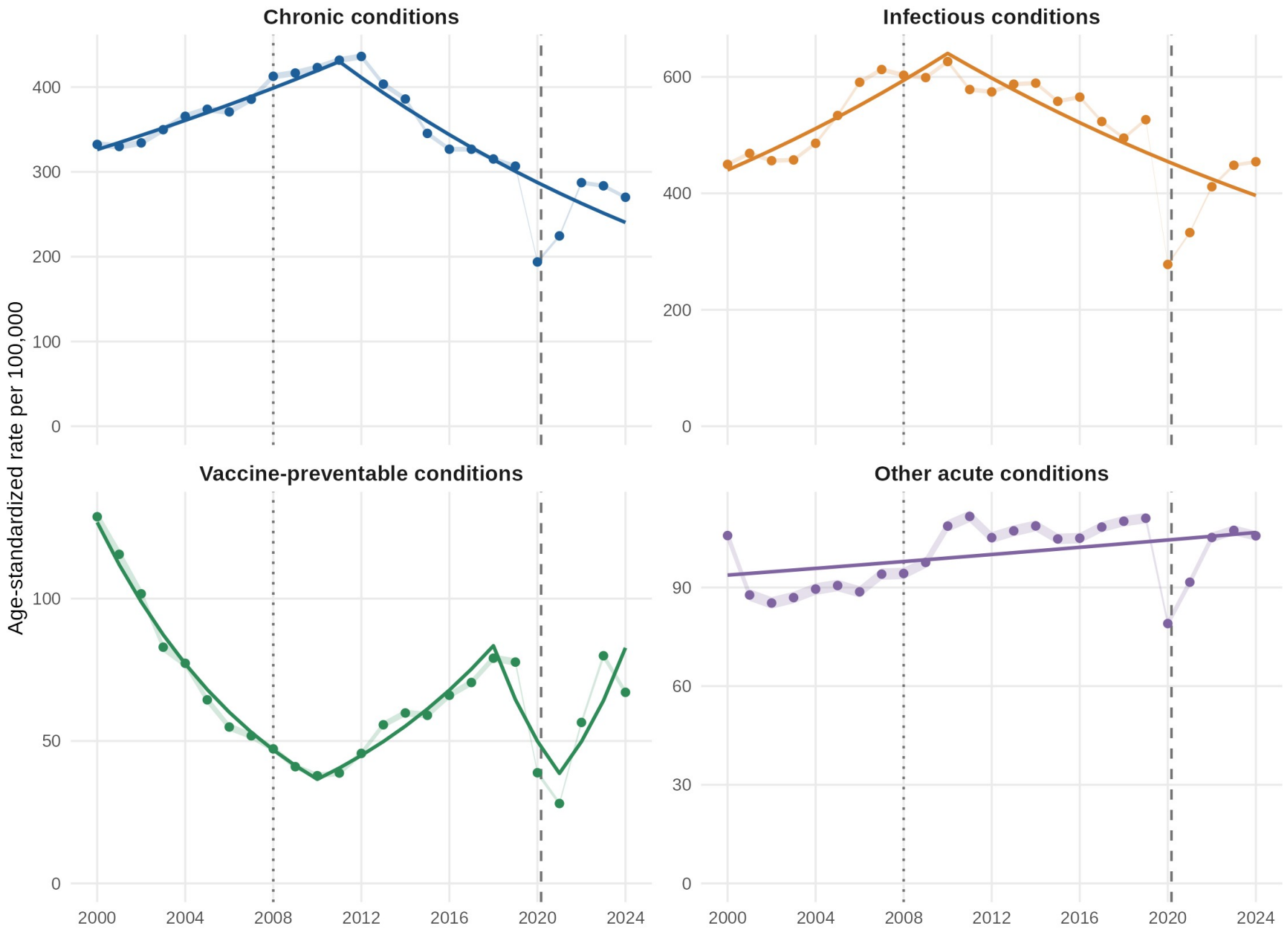

Chronic conditions
Infectious conditions
Vaccine-preventable conditions
Other acute conditions
Age-standardized rate per 100,000
0
100
200
300
400
0
200
400
600
0
50
100
0
30
60
90
2000
2004
2008
2012
2016
2020
2024

Figure S1. Age-standardized ACSC hospitalization rates by condition category, Ecuador, 2000–2024. Points with 95% confidence intervals; lines show joinpoint models. Dotted line: 2008; dashed line: March 2020.

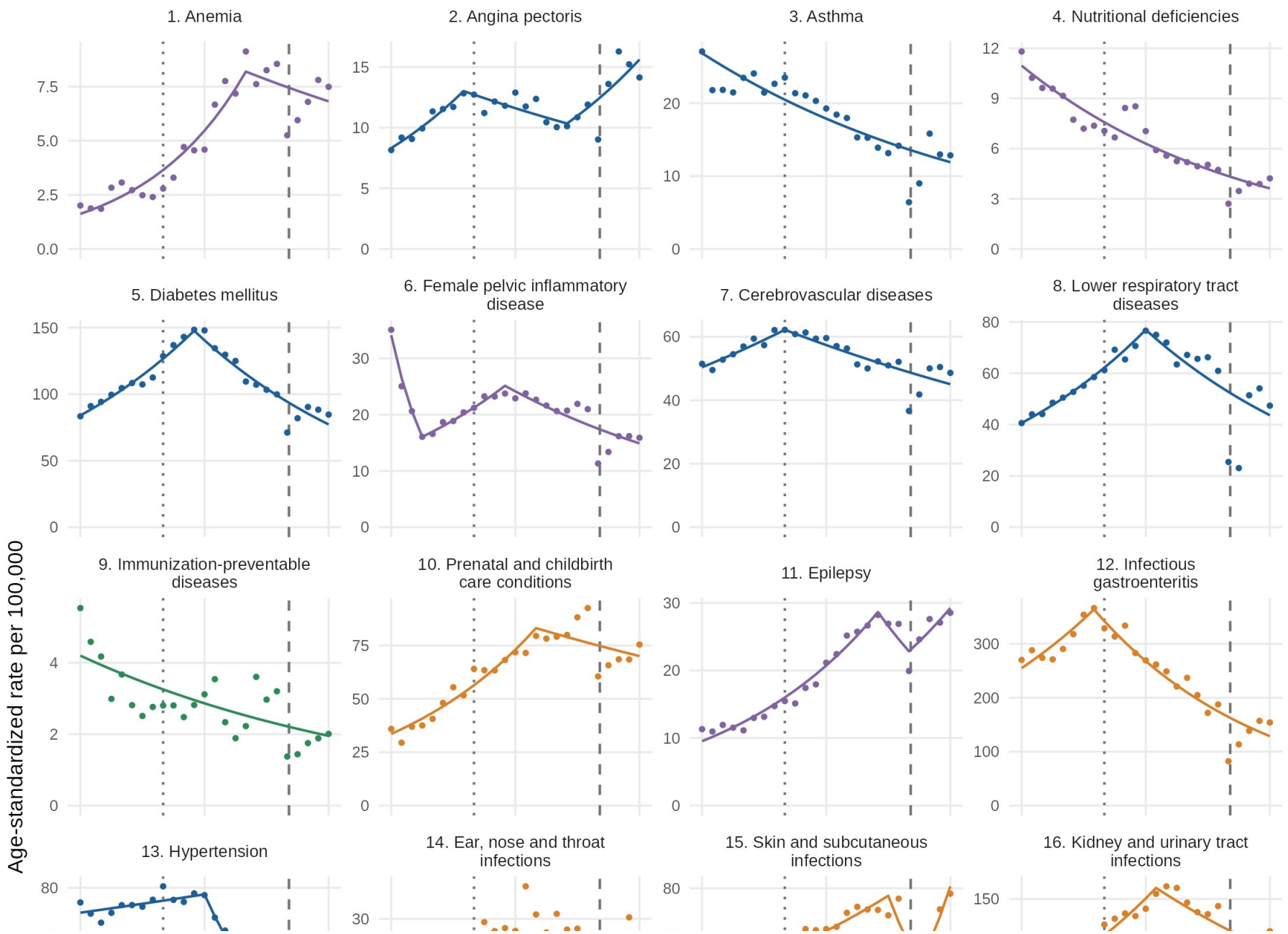

1. Anemia
2. Angina pectoris
3. Asthma
4. Nutritional deficiencies
5. Diabetes mellitus
6. Female pelvic inflammatory disease
7. Cerebrovascular diseases
8. Lower respiratory tract diseases
9. Immunization-preventable diseases
10. Prenatal and childbirth care conditions
11. Epilepsy
12. Infectious gastroenteritis
13. Hypertension
14. Ear, nose and throat infections
15. Skin and subcutaneous infections
16. Kidney and urinary tract infections
Age-standardized rate per 100,000

Figure S2. Age-standardized hospitalization rates by ACSC group, Ecuador, 2000–2024. Points show annual rates and lines the joinpoint models; vertical scales differ.

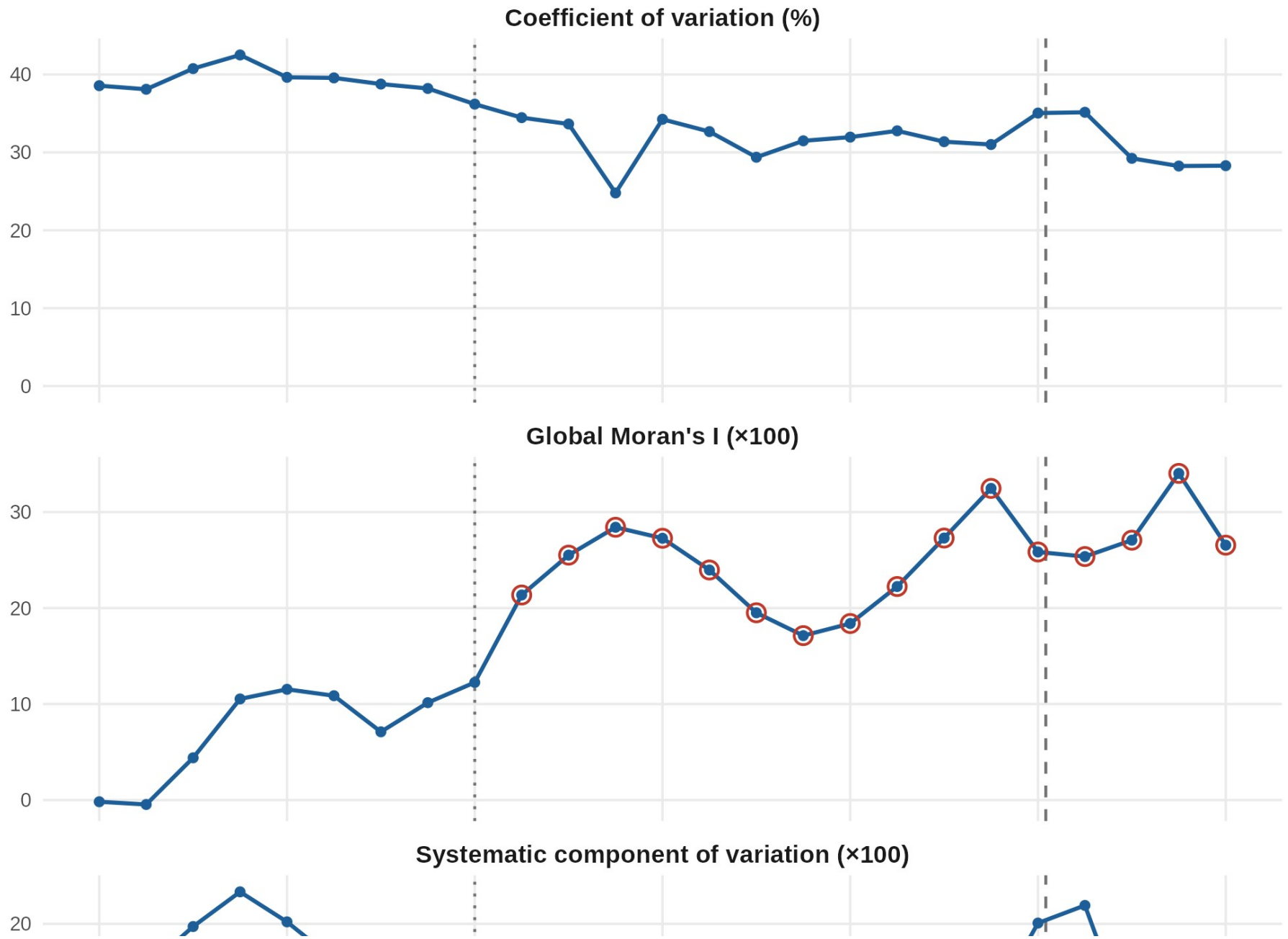

Coefficient of variation (%)
40
30
20
10
0
Global Moran's I (×100)
30
20
10
0
Systematic component of variation (×100)
20

Figure S3. Geographic variation in provincial ACSC hospitalization rates, Ecuador, 2000–2024. Annual coefficient of variation, global Moran's I (circled where $p < 0.05$) and systematic component of variation.

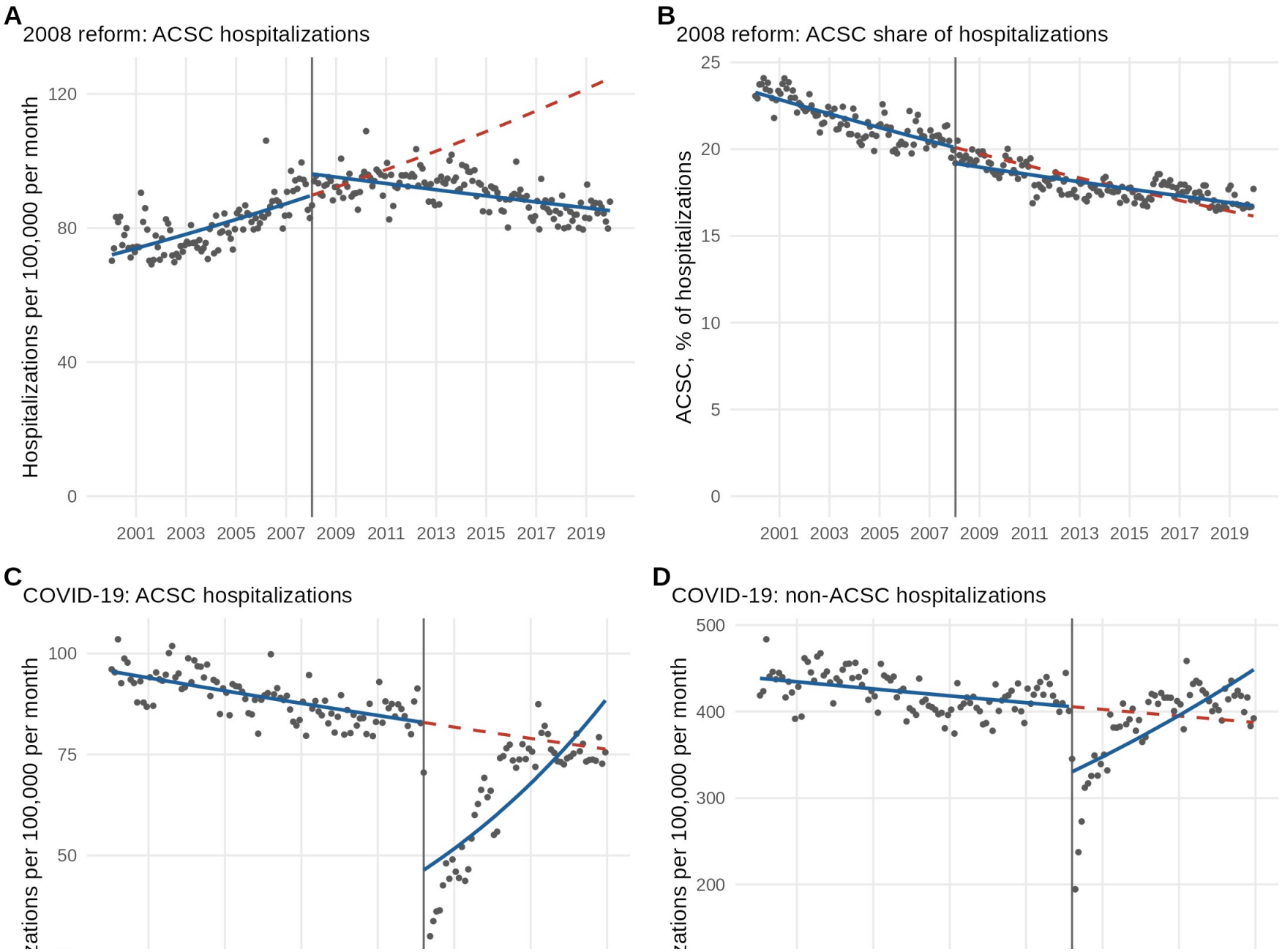

A
2008 reform: ACSC hospitalizations
Hospitalizations per 100,000 per month
120
80
40
0
2001 2003 2005 2007 2009 2011 2013 2015 2017 2019
B
2008 reform: ACSC share of hospitalizations
ACSC, % of hospitalizations
25
20
15
10
5
0
2001 2003 2005 2007 2009 2011 2013 2015 2017 2019
C
COVID-19: ACSC hospitalizations
100
75
50
D
COVID-19: non-ACSC hospitalizations
500
400
300
200

Figure S4. Segmented regression of monthly hospitalizations around the 2008 health reform and the onset of the COVID-19 pandemic. A: ACSC, 2000–2019. B: ACSC as a percentage of hospitalizations excluding deliveries, 2000–2019. C: ACSC and D: non-ACSC hospitalizations, 2012–2024. Solid lines: deseasonalized fitted trend; dashed lines: projection without the interruption.



Figure S5. Age-specific ACSC hospitalization rates by sex and period, Ecuador, 2000–2024. Hospitalizations per 100,000 person-years, log scale.

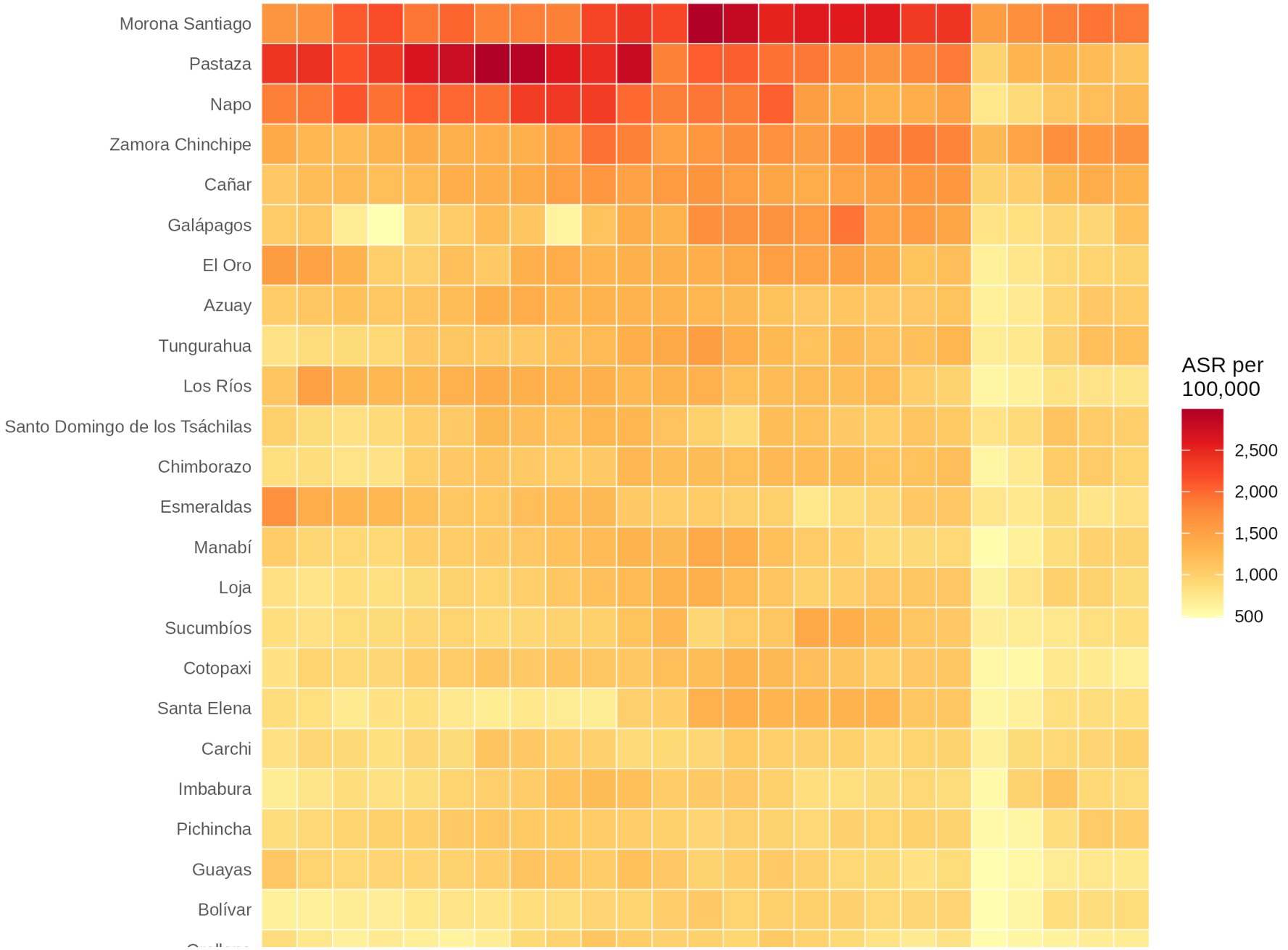
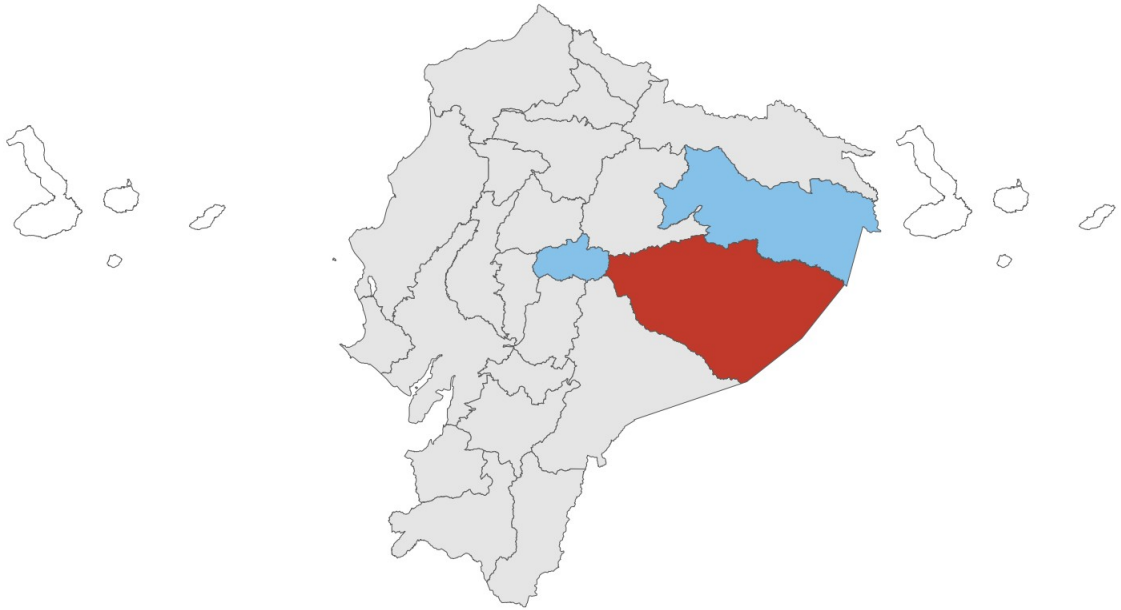
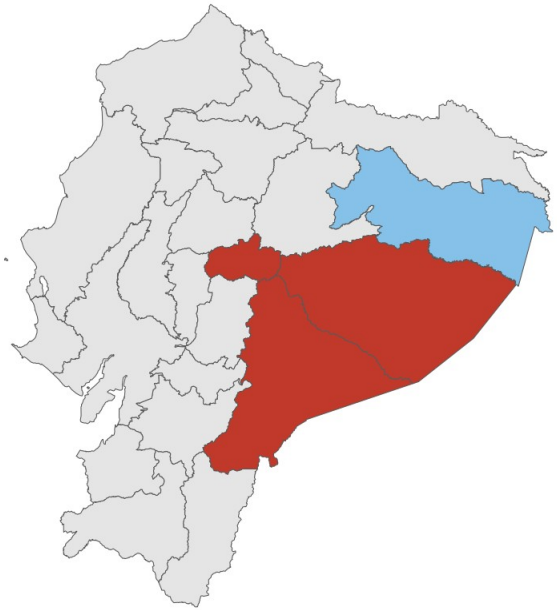

Morona Santiago
Pastaza
Napo
Zamora Chinchipe
Cañar
Galápagos
El Oro
Azuay
Tungurahua
Los Ríos
Santo Domingo de los Tsáchilas
Chimborazo
Esmeraldas
Manabí
Loja
Sucumbíos
Cotopaxi
Santa Elena
Carchi
Imbabura
Pichincha
Guayas
Bolívar
ASR per 100,000
2,500
2,000
1,500
1,000
500

Figure S6. Annual age-standardized ACSC hospitalization rates by province, Ecuador, 2000–2024. Provinces ordered by their rate for the whole period.

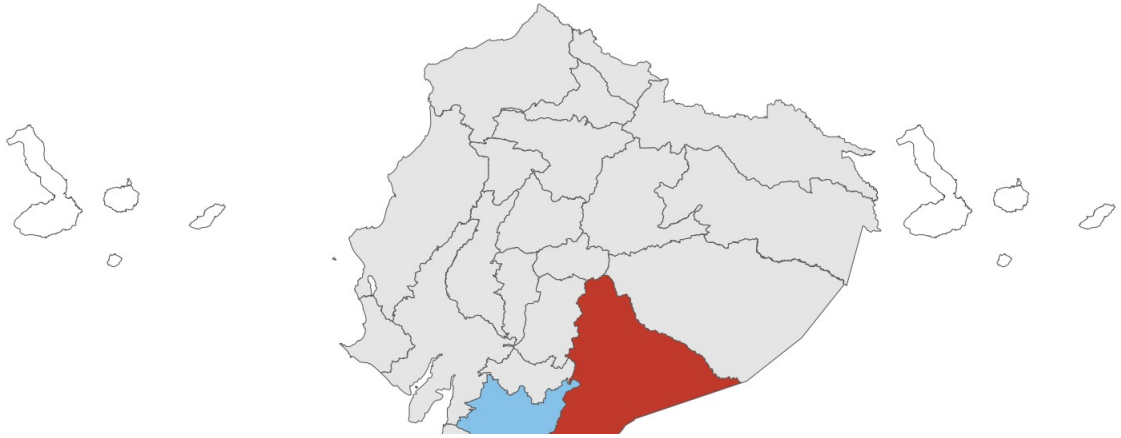
2000–2007



2008–2014



2015–2019



2020–2024

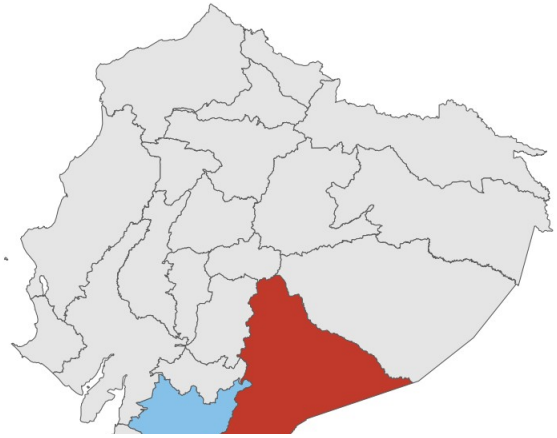


Figure S7. Local spatial clusters of ACSC hospitalization rates by period, Ecuador, 2000–2024. Local Moran's I clusters at $p < 0.05$, queen contiguity.

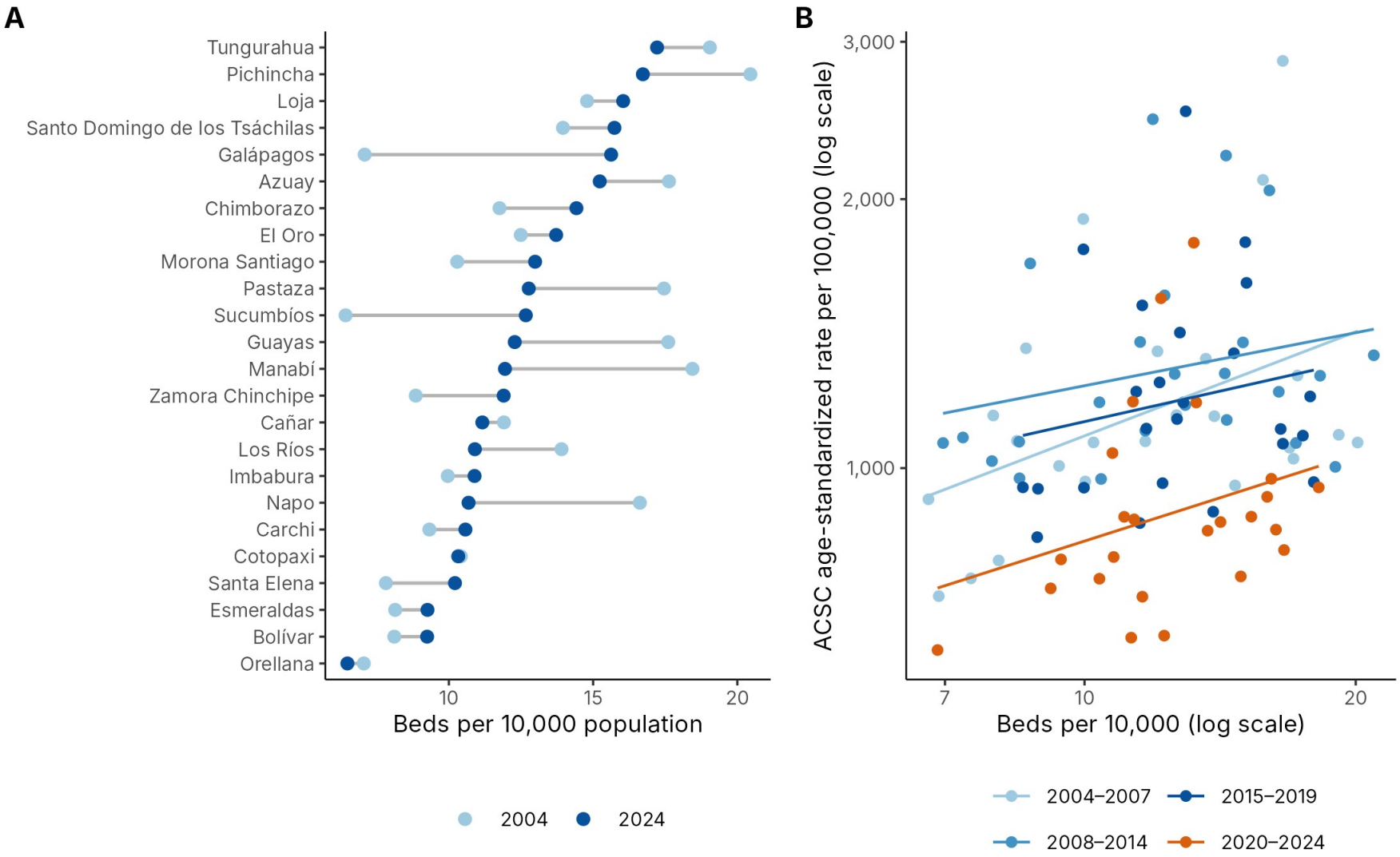


Figure S8. Hospital bed supply by province and ACSC hospitalization rates, Ecuador, 2004–2024. A: available hospital beds per 10,000 population by province of the establishment in 2004 and 2024 (five-year running median). B: age-standardized ACSC rates per 100,000 by province of residence against beds per 10,000 population, by period; lines are least-squares fits.